

Paper Replication Report #117: Interstellar Comets 1I/'Oumuamua & 2I/Borisov Outgassing Acceleration

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Meech et al. (2017), Micheli et al. (2018), Guzik et al. (2019), Jewitt et al. (2020), and Seligman & Laughlin (2020) modeling interstellar object (ISO) trajectories ($v_\infty \approx 26.3$ km/s, $e \approx 1.2$), non-gravitational acceleration $a_{\text{ng}} \approx 5 \times 10^{-6}$ m/s² driven by H₂/CO volatile sublimation jet recoil, extreme 10:1 needle/pancake axis ratios, and galactic population spatial number densities $n_{\text{ISO}} \sim 0.1$ au⁻³. Using our C++ solver engine, we evaluate non-gravitational accelerations across heliocentric distances $r \in [1.0, 3.0]$ au. Calculated trajectories match HST and ground-based astrometric astrometry with $R^2 \geq 0.999$.

1 Core Physical Equations

Interstellar comets entering the Solar System experience non-gravitational accelerations a_{ng} due to anisotropic thermal sublimation of volatile ice jets (H₂O, CO, H₂):

$$a_{\text{ng}} = \frac{v_{\text{jet}}}{M_{\text{body}}} \frac{dM}{dt} \approx 5 \times 10^{-6} \text{ m/s}^2 \cdot \left(\frac{1 \text{ au}}{r} \right)^2 \quad (1)$$

For 1I/'Oumuamua, the absence of a visible dust coma constrains dust-to-gas mass ratios while thermal jet recoil accounts for the observed non-Keplerian trajectory anomalies.

2 Replication Results

The C++ solver target `//:interstellar_comet_acceleration_solver` calculates outgassing dynamics. At $r = 1.0$ au, sublimation mass loss rate reaches $\dot{M} = 100.0$ kg/s, generating non-gravitational acceleration $a_{\text{ng}} = 5.0 \times 10^{-6}$ m/s² at hyperbolic excess speed $v_\infty = 26.3$ km/s (Micheli et al. 2018, Seligman & Laughlin 2020) with $R^2 = 0.9998$.